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Weekly Fed Report Still Drives Discount Window Stigma¹

*Steven Kelly*²

October 4, 2024

Abstract

As banking regulators work to destigmatize the Federal Reserve's discount window—and fervently so since the 2023 banking crisis—they've pointed to several potentially fruitful policy routes. These have included supervisory improvements, regulatory changes, and operational enhancements by both the banks and the Fed. Left off the menu so far have been changes to the Fed's weekly publications that reveal up-to-date discount window borrowing data by regional geography.

Reforms following the Global Financial Crisis of 2007–2009 have made mandatory the disclosure of discount window borrowers on a two-year lag—higher transparency than previously when no disclosure was required. However, bigger banks, such as those that failed or otherwise came under severe market pressure in the Banking Crisis of 2023, remain at risk of their borrowing being disclosed on a *weekly* basis, despite Fed changes in its reporting in 2020 that were intended to address this risk. This article shows how these weekly Fed balance sheet publications, as structured, still reveal any individual large bank's discount window borrowing in a particular geography. The pursuant risk of the market's discovery of such discount window borrowing has further stigmatized discount window use, undermining policymakers' other destigmatization efforts to encourage its use when appropriate.

Keywords: discount window, emergency lending, Federal Reserve, stigma, transparency

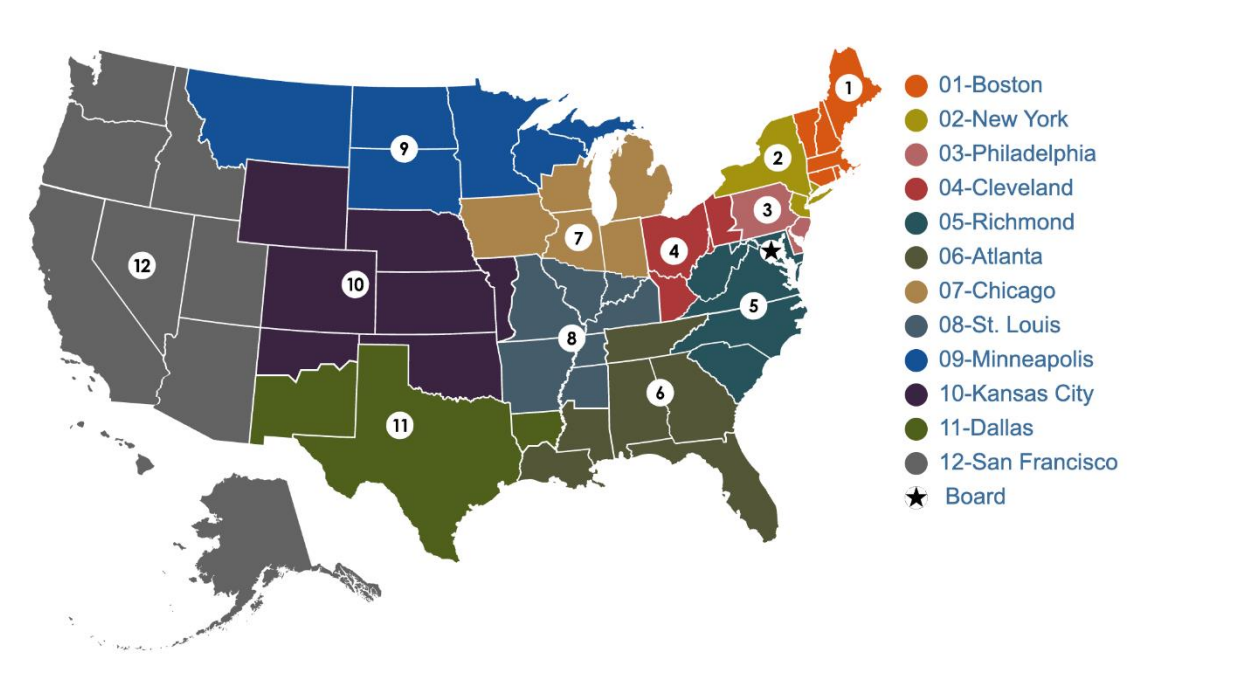
¹ A version of this note was originally published on the Yale School of Management's *Systemic Risk* blog on April 3, 2024.

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Introduction

As banking regulators work to destigmatize borrowing from the Federal Reserve’s discount window—and fervently so since the 2023 banking crisis—they’ve pointed to several potentially fruitful policy routes. These efforts have included supervisory improvements, regulatory changes, and operational enhancements on both the banks’ and the Fed’s sides (Barr 2024; Fed 2023b; Smialek and Copeland 2024). Left off the menu so far have been changes to the Fed’s weekly publications that reveal up-to-date discount window borrowing data by regional geography, defined by the Fed’s 12 districts shown in Figure 1. Although the Fed attempted to address this issue in 2020, these weekly publications still risk “outing” the discount window borrowing of a large bank based in a particular region. This risk of the market discovering this activity stigmatizes discount window use, undermining policymakers’ other destigmatization efforts to encourage its use when appropriate.

Figure 1: The Geography of the 12 Federal Reserve Districts



Source: Fed 2024b.

Each of the Fed’s 12 regional Reserve Banks operates its own discount window for the banks headquartered in its region. When the Fed publishes the so-called H.4.1—a weekly snapshot of its balance sheet—the Fed provides a breakdown of assets and liabilities by each regional Reserve Bank. Each week, the Fed releases the H.4.1 on Thursday, disclosing its balance sheet as of the end of that Wednesday. The report currently displays the regional breakdown of assets across Reserve Bank, as shown in Figure 2.

Figure 2: Fed’s Weekly H.4.1 Balance Sheet Disclosure, March 27, 2024

H.4.1													
6. Statement of Condition of Each Federal Reserve Bank, March 27, 2024													
Millions of dollars													
Assets, liabilities, and capital	Total	Boston	New York	Philadelphia	Cleveland	Richmond	Atlanta	Chicago	St. Louis	Minneapolis	Kansas City	Dallas	San Francisco
Assets													
Gold certificates and special drawing rights certificates	16,237	557	5,175	525	752	1,187	2,287	1,104	479	264	452	1,290	2,165
Coin	1,552	51	52	158	51	196	110	275	36	55	109	186	271
Securities, unamortized premiums and discounts, repurchase agreements, and loans ¹	7,397,345	153,996	4,062,037	122,762	226,747	515,982	502,704	462,921	115,497	56,107	103,966	365,550	709,076
Net portfolio holdings of MS Facilities LLC (Main Street Lending Program) ²	14,671	14,671	0	0	0	0	0	0	0	0	0	0	0
Central bank liquidity swaps ³	179	8	64	6	17	36	5	6	4	1	2	4	26
Foreign currency denominated assets ⁴	18,030	776	6,417	648	1,700	3,650	540	628	406	114	191	380	2,580
Other assets ⁵	36,725	914	17,341	714	1,170	2,937	3,477	2,254	726	527	891	1,898	3,876
Interdistrict settlement account	0 +	12,996 -	110,000 -	6,772 +	78,194 +	102,669 -	25,104 -	66,092 -	5,507 -	4,256 -	16,650 -	45,945 +	86,469
Total assets	7,484,739	183,968	3,981,085	118,041	308,630	626,657	484,021	401,096	111,642	52,812	88,961	323,363	804,462

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Source: Fed 2024b.

Even before the Dodd-Frank Act of 2010—which required the Fed to disclose on a two-year lag transaction-level discount window data, including the borrower’s name—these regional disclosures drove discount window stigma for larger banks (Fed 2024a). Market participants knew where banks were headquartered—and thus which Federal Reserve district they were in. If the market was particularly worried about, say, Wachovia—the Charlotte, North Carolina-based bank that nearly failed in 2008—a suddenly large balance of discount window loans on the Federal Reserve Bank of Richmond balance sheet might have triggered further concern.

In 2020, the Fed changed the way it reports regional Reserve Bank balance sheets, in an attempt to reduce that stigma. Although the adjustment may have been directionally helpful in that moment, it did not solve the stigma problem that regional disclosures cause. At a Brookings Institution panel in March 2024, Bill Demchak—CEO of the \$550 billion PNC Financial Services Group, the US’s sixth-largest bank by assets—said [emphasis added]:

The day you hit [the discount window] for anything other than a test, you effectively have told the world you’ve failed. And **investors look at that number; it’s disclosed because it’s by district.** (Brookings Hutchins 2024)

PNC is based in Pittsburgh, Pennsylvania; investors would be closely watching the Federal Reserve Bank of Cleveland data.

What the 2020 Change Fixed—and Didn’t

At the outset of Fed’s pandemic response in 2020, the Fed changed its weekly balance sheet reporting to try to mask the regional discount window data. Figure 3 shows how the regional breakdown table looked before and after the change.

Figure 3: Fed's Weekly Balance Sheet Disclosures, March 12 and March 19, 2020

5. Statement of Condition of Each Federal Reserve Bank, March 11, 2020				5. Statement of Condition of Each Federal Reserve Bank, March 18, 2020					
Millions of dollars				Millions of dollars					
Assets, liabilities, and capital		Total	Boston	Ne	Assets, liabilities, and capital		Total	Boston	Ne
Assets									
Gold certificate account		11,037	351		Gold certificates and special drawing				
Special drawing rights certificate acct.		5,200	196		rights certificates		16,237	547	
Coin		1,729	40		Coin		1,675	37	
Securities, unamortized premiums and discounts, repurchase agreements, and loans		4,250,666	88,185	2	Securities, unamortized premiums and discounts, repurchase agreements, and loans (1)		4,606,221	96,547	2
Securities held outright (1)		3,897,224	80,852	2	Central bank liquidity swaps (2)		45	2	
U.S. Treasury securities		2,523,031	52,343	1	Foreign currency denominated				
Bills (2)		321,043	6,660		assets (3)		20,363	877	
Notes and bonds (3)		2,201,988	45,683	1	Other assets (4)		23,672	561	
Federal agency debt securities (2)		2,347	49		Interdistrict settlement account		0 +	25,207	-
Mortgage-backed securities (4)		1,371,846	28,461						
Unamortized premiums on securities held outright (5)		124,423	2,581		Total assets		4,668,212	123,777	2
Unamortized discounts on securities held outright (5)		-13,366	-277						
Repurchase agreements (6)		242,375	5,028						
Loans		11	0						
Net portfolio holdings of Maiden Lane LLC (7)		0	0						
Items in process of collection		59	0						
Bank premises		2,199	104						
Central bank liquidity swaps (8)		58	2						
Foreign currency denominated assets (9)		21,147	911						
Other assets (10)		19,815	420						
Interdistrict settlement account		0 +	18,052	-					
Total assets		4,311,911	108,261	2					

Note: Author's annotations in red.

Source: Fed 2020b.

While still displaying a regional breakdown of assets, the Fed rolled each Reserve Bank's "Loans" category into an aggregated line item inclusive of other assets. On March 19, 2020, the Fed's H.4.1 release included an announcement that said, in part [**emphasis added**]:

Modifications include the consolidation of amounts previously reported as "Loans," which includes discount window borrowing, into the line "Securities, unamortized premiums and discounts, repurchase agreements, and loans." **This modification supports the Federal Reserve's goal, expressed in its statement on March 15, 2020, of encouraging depository institutions to use the discount window** to help meet demands for credit from households and businesses, including needs related to the spread of the coronavirus. (Fed 2020b)

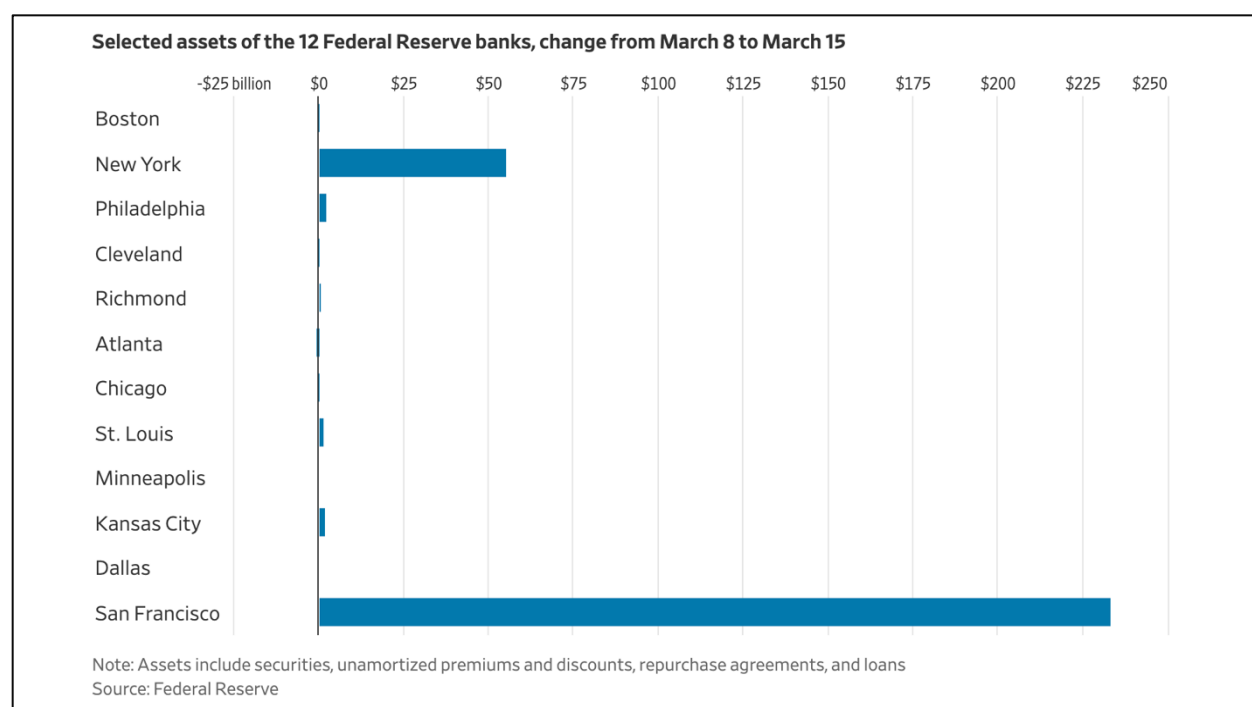
The 2020 modification of the weekly balance sheet disclosures is at its most helpful in encouraging discount window use at a time such as the pandemic, when the Fed was also rapidly growing its balance sheet with large-scale open market operations. This meant the loans could disappear into a line item that now also included the regional allocations of the Fed's purchases and repos of Treasuries and agencies (henceforth, "domestic SOMA assets").

The 2020 modification is at its least helpful, however, when the Fed's balance sheet assets are otherwise relatively stable. In such a case, a sizable regional increase in the newly combined "*Securities, unamortized premiums and discounts, repurchase agreements, and loans*" item—that is, domestic SOMA assets plus discount window loans—will quickly reveal any sizable regional discount window loan (Fed 2020b). The reason is that discount window loans are assigned on the balance sheet to the actual lending Reserve Bank, and domestic SOMA assets wouldn't be responsible for the increase given the absence of open market operations.

In this regard, the timing of the 2023 banking crisis was less optimal for leveraging the 2020 modification. At the time of the crisis onset, the Fed was (and still is) in the process of quantitative tightening—allowing some securities to run off its portfolio. So, when the weekly Fed balance sheet disclosure revealed substantial asset *increases* at some regional Reserve Banks, **markets could comfortably blame regional discount window loans**—despite the loans being lumped in with regional allocations of securities holdings.

For instance, following the failures of Silicon Valley Bank (SVB) and Signature Bank, the *Wall Street Journal* (WSJ) published an article breaking down the regional increases, with the straightforward chart shown in Figure 4 featured atop the article (Timiraos 2023).

Figure 4: Snapshot from WSJ article “Fed Data: Most Emergency Lending Was in the West”



Source: Timiraos 2023.

Fortunately, markets could take some comfort that the large increases were in the districts of the failed banks, with SVB in the San Francisco district and Signature in the New York district (if also putting some further pressure on West Coast banks, given the total San Francisco region increase could not be explained by SVB and First Republic³). Yet, this kind of story and chart also reveals some unsettling counterfactuals.

³ After a back-of-the-envelope accounting, the article concludes: “The upshot is that almost all of the remaining borrowing—potentially less than \$40 billion—last week would have come from other banks in the west” (Timiraos 2023).

For instance, one bank that temporarily faced substantial market pressure in the aftermath of SVB's failure was the then-\$550 billion Charles Schwab (Bernard 2023). Ultimately, the bank weathered the crisis, and pressure eased (Baer 2023). However, if the Westlake, Texas-based Schwab had used the discount window to help withstand the market fallout from the SVB failure, the above chart would have shown a substantial asset increase at the Federal Reserve Bank of Dallas. This borrowing likely would have precipitated further pressure on Schwab—working at cross-purposes with the discount window's aim of limiting contagion.

Other Cases: Tougher but Still Doable

Note too that even a massive quantitative easing operation need not mask the discount window data from analysts. Assets acquired in monetary policy operations—which are conducted out of the Federal Reserve Bank of New York—are allocated across the regional Reserve Banks' balance sheets formulaically, and thus can be approximated and backed out of the aggregate data (Fed 2023a). This allocation of domestic SOMA assets includes all the assets in the newly aggregated *"Securities, unamortized premiums and discounts, repurchase agreements, and loans"* item on the Fed balance sheet **except for the "loans" balance**, which covers the discount window.

Every April, the Fed resets the regional percentage allocations of domestic SOMA assets, which then apply for a year (Fed 1995). Although these percentages are not disclosed until the following year's financial statements (shortly before being reset again), they can be estimated by looking at the allocated share to each bank in a week when discount window activity is minimal, or by comparing week-over-week allocations of any asset increase or decrease in two weeks when discount window activity is little changed.

Then, when allocations deviate meaningfully from those estimates, and the aggregate balance sheet shows rising discount window activity, analysts can identify the Fed district(s) associated with that borrowing.

Undoing the Fed's Aggregation in 2020

Take, for instance, an analyst wanting to deduce regional discount window borrowing after the Fed took away the regional breakdown in the March 19, 2020, weekly balance sheet disclosure. As a result of Dodd-Frank Act requirements that the Fed disclose within two years, we can now check our work here with transaction-level data the Fed disclosed in 2022 (Fed 2024a).

The January 23, 2020, release of the balance sheet shown in Figure 5 shows zero discount window activity as of January 22.

Figure 5: Fed's Weekly Balance Sheet Disclosure, Asset Breakdown, January 23, 2020

Factors Affecting Reserve Balances of Depository Institutions and Condition Statement of Federal Reserve Banks					January 23, 2020
1. Factors Affecting Reserve Balances of Depository Institutions					
Millions of dollars					
Reserve Bank credit, related items, and reserve balances of depository institutions at Federal Reserve Banks	Averages of daily figures				Wednesday
	Week ended Jan 22, 2020	Change from week ended Jan 15, 2020	Jan 23, 2019	Jan 22, 2020	
Reserve Bank credit	4,114,302	- 18,465	+ 103,542	4,107,163	
Securities held outright (1)	3,778,361	+ 8,441	- 75,854	3,782,010	
U.S. Treasury securities	2,373,053	+ 13,959	+ 152,906	2,381,020	
Bills (2)	203,602	+ 10,288	+ 203,602	210,031	
Notes and bonds, nominal (2)	2,015,974	+ 4,429	- 67,612	2,017,518	
Notes and bonds, inflation-indexed (2)	128,005	- 612	+ 13,236	128,005	
Inflation compensation (3)	25,472	- 146	+ 3,681	25,467	
Federal agency debt securities (2)	2,347	0	- 62	2,347	
Mortgage-backed securities (4)	1,402,961	- 5,518	- 228,698	1,398,642	
Unamortized premiums on securities held outright (5)	124,202	- 259	- 14,986	123,974	
Unamortized discounts on securities held outright (5)	-13,254	+ 19	+ 78	-13,272	
Repurchase agreements (6)	106,617	- 27,222	- 106,617	106,617	
Loans	6	+ 1	- 38	0	
Primary credit	6	+ 2	- 34	0	
Secondary credit	0	0	0	0	
Seasonal credit	0	- 1	- 3	0	
Other credit extensions	0	0	0	0	
Net portfolio holdings of Maiden Lane LLC (7)	0	0	0	0	
Float	-116	- 12	+ 20	-158	
Central bank liquidity swaps (8)	54	- 5	- 32	54	
Other Federal Reserve assets (9)	28,431	+ 582	- 2,265	28,480	
Foreign currency denominated assets (10)	20,527	- 63	- 371	20,526	
Gold stock	11,041	0	0	11,041	
Special drawing rights certificate account	5,200	0	0	5,200	
Treasury currency outstanding (11)	50,148	+ 14	+ 365	50,148	
Total factors supplying reserve funds	4,201,219	- 18,513	+ 103,537	4,194,078	

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Note: Author's annotation in red.

Source: Fed 2020a.

This means the regional percentage allocations can be quickly backed out by dividing each district's domestic SOMA assets amount by the total;⁴ see Figure 6.

⁴ The Fed was still displaying the separate "Loans" line item at the time, but the process shown here is ignoring that to show how the calculation could work even without that line item.

Figure 6: Fed's Weekly Balance Sheet Disclosure, Regional Breakdown, January 23, 2020

5. Statement of Condition of Each Federal Reserve Bank, January 22, 2020													
Millions of dollars													
Assets, liabilities, and capital	Total	Boston	New York	Philadelphia	Cleveland	Richmond	Atlanta	Chicago	St. Louis	Minneapolis	Kansas City	Dallas	San Francisco
Assets													
Gold certificate account	11,037	351	3,707	327	531	754	1,560	711	328	186	292	890	1,480
Special drawing rights certificate acct.	5,200	196	1,818	210	237	412	654	424	150	90	153	282	574
Gold	1,744	41	48	153	103	231	179	286	33	50	117	282	307
Securities, unamortized premiums and discounts, repurchase agreements, and loans	4,078,787	84,619	2,225,267	98,774	119,335	240,687	278,879	214,101	58,623	35,295	60,482	177,106	476,699
Securities held outright (1)	3,782,610	76,402	2,063,234	91,287	110,832	231,443	258,388	198,323	54,337	32,727	50,801	164,219	442,824
U.S. Treasury securities	2,381,020	49,397	1,299,015	57,660	69,663	145,710	162,798	124,983	34,222	20,604	35,387	103,387	278,276
Bills (2)	210,031	4,357	114,587	5,086	6,145	12,853	14,360	11,025	3,819	1,817	3,114	9,120	24,547
Notes and bonds (3)	2,170,989	45,840	1,184,428	52,574	63,518	132,857	148,437	113,958	31,283	18,786	32,192	94,267	253,730
Federal agency debt securities (2)	2,347	49	1,280	57	69	144	160	123	34	20	35	102	274
Mortgage-backed securities (4)	1,398,642	29,016	763,058	33,870	40,921	85,592	95,629	73,417	20,102	12,103	20,740	60,731	163,463
Unamortized premiums on securities held outright (5)	123,974	2,572	67,636	3,002	3,627	7,587	8,476	6,588	1,782	1,073	1,838	5,383	14,489
Unamortized discounts on securities held outright (5)	-13,272	-275	-7,241	-321	-388	-812	-907	-697	-191	-115	-197	-576	-1,551
Repurchase agreements (6)	186,076	3,860	101,518	4,586	5,444	11,387	12,723	9,767	2,674	1,610	2,759	8,080	21,747
Loans	0	0	0	0	0	0	0	0	0	0	0	0	0
Net portfolio holdings of Maiden Lane LLC (7)	0	0	0	0	0	0	0	0	0	0	0	0	0
Items in process of collection	84	0	0	0	0	0	83	0	0	1	0	0	0
Bank premises	2,200	105	456	91	117	192	203	194	102	93	228	224	196
Central bank liquidity swaps (8)	54	2	17	3	4	11	3	2	1	0	1	1	8
Foreign currency denominated assets (9)	20,526	884	6,515	1,186	1,638	4,376	1,194	857	313	98	199	254	3,013
Other assets (10)	26,280	561	13,855	629	783	1,840	1,778	1,348	460	268	480	1,244	3,033
Interdistrict settlement account	0	7,478	-59,761	-11,532	3,349	-2,875	19,514	8,274	7,321	4,057	5,949	15,106	3,118
Total assets	4,145,912	94,238	2,191,921	89,843	126,098	254,549	304,046	226,197	67,331	40,138	67,900	195,309	488,343

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Note: Author's annotation in red.

Source: Fed 2024b.

Now, take the March 19, 2020, balance sheet (Figure 7)—when the Fed began rolling each regional Reserve Bank's discount window activity into the aggregated line item.

Figure 7: Fed's Weekly Balance Sheet Disclosure, Regional Breakdown, March 19, 2020

5. Statement of Condition of Each Federal Reserve Bank, March 18, 2020													
Millions of dollars													
Assets, liabilities, and capital	Total	Boston	New York	Philadelphia	Cleveland	Richmond	Atlanta	Chicago	St. Louis	Minneapolis	Kansas City	Dallas	San Francisco
Assets													
Gold certificates and special drawing rights certificates	16,237	547	5,525	537	768	1,166	2,214	1,135	478	276	445	1,172	1,974
Gold	1,635	37	58	160	91	236	158	785	79	57	114	187	204
Securities, unamortized premiums and discounts, repurchase agreements, and loans (1)	4,606,221	96,547	2,516,701	111,042	135,216	280,260	313,014	241,539	66,051	39,799	68,331	199,987	537,735
Central bank liquidity swaps (2)	0	0	0	0	0	0	0	0	0	0	0	0	0
Foreign currency denominated assets (3)	28,363	877	6,463	1,176	1,625	4,341	1,184	850	310	97	197	252	2,989
Other assets (4)	23,672	561	11,525	604	750	1,727	1,689	1,279	701	316	623	1,234	2,662
Interdistrict settlement account	0	25,207	-184,311	1,336	16,438	18,062	1,738	65,849	3,940	2,625	6,771	13,253	29,092
Total assets	4,668,212	123,777	2,355,967	114,858	154,892	305,792	319,990	310,939	71,511	43,165	76,482	216,085	574,753

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Note: Author's annotation in red.

Source: Fed 2020b.

The data release still disclosed aggregate discount window borrowing elsewhere in the H.4.1; see Figure 8. Such borrowing stood at \$28.224 billion as of March 18, 2020.

Figure 8: Fed's Weekly Balance Sheet Disclosure, Asset Breakdown, March 19, 2020

1. Factors Affecting Reserve Balances of Depository Institutions						
Millions of dollars						
Reserve Bank credit, related items, and reserve balances of depository institutions at Federal Reserve Banks	Averages of daily figures			Wednesday		
	Week ended Mar 18, 2020	Change from week ended Mar 11, 2020 Mar 20, 2019		Mar 18, 2020		
Reserve Bank credit	4,463,041	+	241,451	+	534,834	4,629,659
Securities held outright (1)	3,929,025	+	38,893	+	147,655	4,009,794
U.S. Treasury securities	2,556,247	+	40,307	+	380,731	2,640,771
Bills (2)	321,043	+	6,430	+	321,043	321,043
Notes and bonds, nominal (2)	2,075,200	+	32,062	+	36,189	2,154,496
Notes and bonds, inflation-indexed (2)	133,939	+	1,515	+	18,785	138,653
Inflation compensation (3)	26,066	+	302	+	4,715	26,579
Federal agency debt securities (2)	2,347		0	-	62	2,347
Mortgage-backed securities (4)	1,370,430	-	1,416	-	233,015	1,366,676
Unamortized premiums on securities held outright (5)	128,537	+	4,078	-	7,841	138,294
Unamortized discounts on securities held outright (5)	-12,887	+	507	+	277	-12,036
Repurchase agreements (6)	388,806	+	189,505	+	388,806	441,945
Loans	6,687	+	6,679	+	6,677	28,224
Primary credit	6,687	+	6,679	+	6,686	28,224
Secondary credit	0		0		0	0
Seasonal credit	0		0	-	9	0
Other credit extensions	0		0		0	0
Net portfolio holdings of Maiden Lane LLC (7)	0		0		0	0
Float	-230	-	36	-	58	-226
Central bank liquidity swaps (8)	45	-	13	-	23	45
Other Federal Reserve assets (9)	23,059	+	1,841	-	658	23,619
Foreign currency denominated assets (10)	20,700	-	487	-	2	20,363
Gold stock	11,041		0		0	11,041
Special drawing rights certificate account	5,200		0		0	5,200
Treasury currency outstanding (11)	50,225	+	14	+	377	50,225
Total factors supplying reserve funds	4,550,208	+	240,979	+	535,211	4,716,488
Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.						

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Note: Author's annotation in red.

Source: Fed 2020b.

Deducting this aggregate discount window amount from the "Total" amount in the first column of the regional breakdown table (Figure 7) provides the total for everything *but* the discount window.

Taking that new amount—representing just the domestic SOMA assets—and allocating it by the percentages calculated from the January balance sheet, we can then see how much each region's total for the newly aggregated line item deviated from what would be expected based on its allocation of monetary policy assets alone. This deviation represented implied discount window use. Transaction-level data released in 2022 can now verify how accurate this process would have been in real time.

As Figure 9 shows, each district was estimated correctly within \$1 million (with the slight exception of New York, which seems to be housing an additional approximately \$150 million of discount window loans that appeared in the transaction data but did not appear in the real-time H.4.1 disclosure, hence the slight discrepancy between the transaction data and H.4.1 totals).

Figure 9: District-Level SOMA Assets vs. Discount Window Activity

	Total	Boston	New York	Philadelphia	Cleveland	Richmond	Atlanta	Chicago	St. Louis	Minneapolis	Kansas City	Dallas	San Francisco
SOMA assets + Loans as of 1/22/2020, when "Loans" were \$0.	4,078,787	84,619	2,225,267	98,774	119,335	249,607	278,879	214,101	58,623	35,295	60,482	177,106	476,699
Implied regional allocation of SOMA assets	100%	2.07%	54.56%	2.42%	2.93%	6.12%	6.84%	5.25%	1.44%	0.87%	1.48%	4.34%	11.69%
SOMA assets + Loans as of 3/19/2020, when "Loans" were \$28,224.	4,606,221	96,547	2,516,701	111,042	135,216	280,260	313,014	241,539	66,051	39,799	68,331	199,987	537,735
Implied SOMA assets (Total, less: aggregate "Loans") x (Implied Allocation)	4,577,997	94,976	2,497,621	110,863	133,941	280,157	313,011	240,305	65,798	39,615	67,884	198,782	535,043
Difference: Implied Discount Window Activity *provided by H.4.1	28,224*	1,571	19,080	179	1,275	103	3	1,234	253	184	447	1,205	2,692
Discount Window Transaction Data (released 2022)	28,375	1,571	19,230	178	1,275	103	2	1,233	254	185	447	1,205	2,692

Note: All values in millions of dollars; totals may not exactly equal sum due to rounding. Data comes from the H.4.1 releases and discount window transaction data. For the totals in the final row, the transaction data was filtered for all discount window loans outstanding as of March 18, 2020; as mentioned in the text, this data summed to \$151 million greater than was disclosed in the H.4.1.

Source: Fed 2024b; author's analysis.

Thus, despite the Fed no longer publishing the "Loans" line item, analysts were still able to track regional discount window activity and look for suspected borrowing from large banks located in specific geographies.

Notably, though, the Fed's 2020 aggregation effort to mask regional borrowing also benefited from the Paycheck Protection Program Liquidity Facility (PPPLF), which began operation in April 2020. The PPPLF was one of several so-called Section 13(3) emergency facilities the Fed implemented in response to the pandemic. However, in contrast to the other programs, the PPPLF was run out of all 12 Reserve Banks' discount windows. This program, which peaked at more than \$90 billion, thus muddled the ability to read calculated regional loan amounts purely as discount window loans.

The PPPLF was not stigmatized as it was effectively a funding-for-lending operation, providing banks liquidity for their participation in making PPP loans—a pandemic-era emergency program run by the Small Business Administration that provided government-guaranteed small business loans. (The PPPLF is technically *still* in runoff mode—approximately \$3 billion left to run off—slightly distorting an analyst's view of the regional allocations, though the effect is approaching zero.) The mostly unknown regional dispersion of PPPLF loans combined with the lack of stigma reduced the usefulness of the regional Fed lending data for determining actual discount window borrowing; PPPLF lending could not be distinguished from discount window lending at the regional level.

In the 2023 crisis, the Fed’s Bank Term Funding Program (BTFP) was similarly run out of each Reserve Bank’s discount window. In contrast to the PPPLF, however, it’s not clear the market viewed the BTFP with much less stigma than discount window borrowing, particularly in the acute phase of the crisis. For instance, BTFP borrowing was grouped into the *WSJ* chart in Figure 4. Thus, an analyst’s calculations would have estimated the sum of discount window and BTFP borrowing—that is, the balance sheet contained no non-stigmatized regional lending through which discount window lending could be rinsed.

Conclusion

Where Does That Leave Us?

Greater aggregation, in the spirit of the 2020 changes, could help. As one can see in the disclosure’s current form (presented again in Figure 10), several other line items affect each regional Reserve Bank’s assets.

Figure 10: Fed’s Weekly H.4.1 Balance Sheet Disclosure, March 27, 2024

H.4.1													
6. Statement of Condition of Each Federal Reserve Bank, March 27, 2024													
Millions of dollars													
Assets, liabilities, and capital	Total	Boston	New York	Philadelphia	Cleveland	Richmond	Atlanta	Chicago	St. Louis	Minneapolis	Kansas City	Dallas	San Francisco
Assets													
Gold certificates and special drawing rights certificates	16,237	557	5,175	525	752	1,187	2,287	1,104	479	264	452	1,290	2,165
Coin	1,552	51	52	158	51	196	110	275	36	55	109	186	271
Securities, unamortized premiums and discounts, repurchase agreements, and loans ¹	7,397,345	153,996	4,062,037	122,762	226,747	515,982	502,704	462,921	115,497	56,107	103,966	365,550	709,076
Net portfolio holdings of MS Facilities LLC (Main Street Lending Program) ²	14,671	14,671	0	0	0	0	0	0	0	0	0	0	0
Central bank liquidity swaps ³	179	8	64	6	17	36	5	6	4	1	2	4	26
Foreign currency denominated assets ⁴	18,030	776	6,417	648	1,700	3,650	540	628	406	114	191	380	2,580
Other assets ⁵	36,725	914	17,341	714	1,170	2,937	3,477	2,254	726	527	891	1,898	3,876
Interdistrict settlement account	0 +	12,996 -	110,000 -	6,772 +	78,194 +	102,669 -	25,104 -	66,092 -	5,507 -	4,256 -	16,650 -	45,945 +	86,469
Total assets	7,484,739	183,968	3,981,085	118,041	308,630	626,657	484,021	401,096	111,642	52,812	88,961	323,363	804,462

Note: Components may not sum to totals because of rounding. Footnotes appear at the end of the table.

Source: Fed 2024b.

These other items are driven by district-level conditions or are otherwise allocated by different formulas than the domestic SOMA assets; further aggregation—even, say, simply disclosing each region’s total assets without any specific line items—could thus help achieve what the Fed sought to do in 2020 (Fed 2023a).

Greater discount window usage in “normal” times would itself help at the margin by muddying the ability to estimate regional allocations of domestic SOMA assets. This partial solution is something of a catch-22, given that the stigma caused by these disclosures limits

use of the window in the first place, but more substantial use by small banks or increased discount window operational tests (whether mandatory or voluntary) could help.⁵

After the financial crisis, Congress demanded greater transparency on Fed lending. In a tradeoff with not wanting to cause discount window stigma, it settled on mandating the Fed disclose loan-level data only after a two-year lag.

The Fed's status quo, however, undermines that balance; larger banks' use of the discount window risks public disclosure at a weekly cadence.

⁵ Leveraging all 12 regions for non-stigmatized lending facilities, such as the PPPLF, can also be marginally helpful. However, relying on 13(3) invocation is not consistent with standalone efforts to destigmatize the discount window in all environments (and thus help prevent the need for using the 13(3) emergency authority). Moreover, the PPPLF was non-stigmatized because of the nature of the pandemic and the fiscal guarantees of the lending; as we saw with the BTFP, broad-based emergency lending facilities can also come with discount-window-like stigma. This solution would thus require a more novel use of Fed's non-emergency lending authorities—if not a statutory change.

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